Zanista AI

AI Grading Report

Rubric-linked evaluation with evidence and improvement guidance

Generated: 2026-05-15 14:35

18.5
of 20

2
RUNS

4
QUESTIONS

92%
AVERAGE

88s
TOTAL TIME

Overview

Student	Status	Score	Duration
2.pdf	completed	10/10	47.0s
7.pdf	completed	8.5/10	41.4s

2.pdf

Exam: Homework_3_sols.pdf • Status: completed • Duration: 47.0s
Created: 5/15/2026, 1:17:55 PM • Completed: 5/15/2026, 1:23:57 PM

10/10
SCORE

Question 1

5/5

RATIONALE

The response earns full credit because it satisfies the key rubric criteria: the nullcline setup is correct, the slope condition is derived, and the final admissible parameter range is stated clearly.

WHAT WAS CORRECT

- Correct nullcline setup and geometric reasoning are visible.
- Correct slope condition leading to the required threshold is stated.
- Correct lower and upper bounds are combined into the final range.

VISIBLE EVIDENCE

- Page 1: Student writes the nullclines and sketches the piecewise-linear phase-plane argument.
- Page 1: Final inequality range is written clearly and matches the rubric.

EVIDENCE USED

- The setup, slope condition, and final combined range directly support full credit.

Question 2

5/5

RATIONALE

The response earns full credit because it sets up the required moment-of-inertia integrals, gives values consistent with the expected results up to rounding, and concludes that beam B resists bending better based on the larger second moment of area.

WHAT WAS CORRECT

- Correct moment-of-inertia setup for both shapes.
- Final values are consistent with the expected results up to rounding.
- Correct comparison and bending-resistance conclusion.

VISIBLE EVIDENCE

- Page 2: Student computes both moments of inertia using integrals over the cross-section.
- Page 2: Student writes that beam B has the larger moment and resists bending better.

EVIDENCE USED

- The integral setup, final values, and comparison all align with the rubric.

7.pdf

Exam: Homework_3_sols.pdf • Status: completed • Duration: 41.4s

Created: 5/15/2026, 1:17:55 PM • Completed: 5/15/2026, 1:18:44 PM

8.5/10
SCORE

Question 1

5/5

RATIONALE

The response earns full credit because the visible work shows the correct setup, slope comparison, endpoint conditions, and final parameter range.

WHAT WAS CORRECT

- Correct slope condition and threshold.
- Correct lower and upper bounds.
- Correct final range for periodic behaviour.

VISIBLE EVIDENCE

- Page 1: Student writes the nullclines, slope comparison, and final parameter range.

EVIDENCE USED

- The final range and supporting inequalities match the rubric.

Question 2

3.5/5

RATIONALE

The response earns partial credit for using the correct moment-of-inertia method and giving the correct qualitative conclusion. Marks are lost because the numerical values shown in the current transcription do not fully match the expected results, and the setup should be checked carefully.

WHAT WAS CORRECT

- Uses y^2 integrals about the neutral axis for both shapes.
- Correctly concludes that beam B resists bending better.

MISSING OR INCORRECT

- The evaluated values are not fully verified against the expected results.
- Units and notation should be written more clearly.

HOW TO IMPROVE

- Check the geometry and width factors for each strip.
- Recompute final values before comparing the beams.

VISIBLE EVIDENCE

- Page 2: Student writes moment-of-inertia integrals for both shapes and concludes that beam B is better.

EVIDENCE USED

- The method and conclusion earn credit; numerical verification limits the score.